OptiDerma: Multimodal Dermatology AI Decision-Support Platform

Complete Technical Architecture, Algorithmic Deep-Dive, Live Demonstration Guide & University Thesis Presentation Report

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Document Version: 3.4.0 (Final Release & Presentation Baseline)

System Architecture: Next.js 15 (App Router, Turbopack, TailwindCSS) + FastAPI 0.115 (Python 3.12, PyTorch 2.6, EfficientNet-B4, U-Net) + SQLite (WAL) / Supabase Cloud PostgreSQL

Verification Audit Baseline: 67/67 Backend PyTorch/Agent Tests Passed | 0 TypeScript Errors | Turbopack Production Build Passed | 9 Live Screenshots Captured

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Chapter 1: Project Overview

1.1 What is OptiDerma?

OptiDerma is an enterprise-grade, multimodal artificial intelligence decision-support platform engineered to assist primary care physicians, triage nurses, and clinical practitioners in the pre-biopsy screening, risk stratification, and patient education of cutaneous skin lesions.

Unlike traditional computer vision models that merely output an uncalibrated diagnostic label, OptiDerma combines:

1. **Multi-Task Deep Learning:** Simultaneous 7-class lesion classification and pixel-level lesion boundary segmentation from an EfficientNet-B4 backbone.

2. **Clinical Explainable AI (XAI):** Artifact-resistant Grad-CAM saliency heatmaps coupled with morphological hair removal (DullRazor) and quantitative geometric morphometrics (asymmetry index, compactness, area, perimeter).

3. **Evidence-Grounded Agentic RAG:** An 8-node stateful clinical orchestrator that routes patient inquiries to 34 peer-reviewed clinical practice guidelines (JAAD, BJD, AAD) with strict zero-hallucination guarantees and traceable bracketed citations (`[CIT-X]`).

4. **Resilient Multimodal Access:** Spoken voice-turn pipelines (Whisper STT + ElevenLabs/gTTS) and verified tertiary hospital referrals (Care Connect) backed by dual-store persistence (SQLite WAL + Supabase PostgreSQL).

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■ OptiDerma Decision-Support System ■
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■ Computer Vision ■ ■ Explainable AI ■ ■ Agentic RAG ■ ■ Care Connect ■
■ EfficientNet-B4 ■ ■ Grad-CAM Energy ■ ■ 34 Guidelines ■ ■ Verified Clinics ■
■ 7-Class + U-Net ■ ■ DullRazor Filter ■ ■ Zero-Hallucin. ■ ■ Patient Referral ■
■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■
```

1.2 The Problem OptiDerma Aims to Address

Cutaneous malignancies—including malignant melanoma, basal cell carcinoma, and squamous cell carcinoma—represent one of the fastest-growing cancer burdens globally. Melanoma accounts for the vast majority of skin cancer deaths, yet when detected at Stage I (localized), the 5-year survival rate exceeds **99%**. If diagnosis is delayed to Stage IV (distant metastasis), 5-year survival plummets below **30%**.

The clinical landscape faces three critical bottlenecks:

1. **Dermatologist Shortages & Triage Bottlenecks:** Specialist wait times often exceed 3 to 6 months. Non-specialist primary care physicians face high diagnostic uncertainty, yielding benign-to-malignant biopsy ratios as high as 15:1.

2. **"Black-Box" AI Distrust:** Conventional deep learning models provide softmax probabilities without explaining *where* the model looked, leaving clinicians unable to verify whether the AI analyzed genuine pigment patterns or spurious artifacts (e.g., overlying hair, skin creases, or camera vignetting).

3. **Hallucinatory Medical Chatbots:** Generative language models frequently invent medical guidelines, fabricate clinical study citations, and fail to ground answers in the patient's objective imaging metrics.

1.3 Intended Users & Clinical Context

- **Primary Care Physicians & General Practitioners (GPs):** Objective pre-referral risk assessment to distinguish urgent lesions needing immediate biopsy from low-risk lesions that can be monitored.

- **Triage Nurses & Teledermatology Technicians:** Fast-fail image quality validation and standardized morphometric measurement acquisition.

- **Dermatology Residents & Medical Trainees:** Interactive educational tool providing side-by-side comparative attributions and literature guideline correlations.

- **Patients (Educational Decision Support):** Structured post-analysis summaries explaining why a doctor's visit is needed without providing self-treatment illusions.

1.4 Objectives & Project Scope

Objective	Engineering Implementation	Scope Boundary
High-Resolution Vision	PyTorch EfficientNet-B4 multi-task architecture fine-tuned on HAM10000	7 discrete classes; macroscopic clinical phone photos restricted
Artifact Resistance	96-kernel directional morphological DullRazor with Telea inpainting	Preprocessor filters hairs up to 9px thickness without erasing pigment
Explainability	Analytical Grad-CAM gradients with Spatial Energy Ratio quantification	Explains model focus; does not replace histopathologic analysis
Guideline Grounding	Hybrid BM25/TF-IDF RAG over 34 peer-reviewed guidelines	Strictly answers indexed dermatology queries; refuses out-of-domain queries
Data Durability	Local SQLite WAL store with cloud Supabase replication	Ensures 100% offline uptime; cloud synchronization when online

1.5 What the System Can and Cannot Currently Do

- **What it CAN do:**
- Audit dermoscopic image quality (blur via Laplacian variance, glare, underexposure).
- Automatically detect and inpaint occluding hair fibers using DullRazor.
- Compute probabilities for 7 lesion types and classify overall risk into High, Moderate, or Low.
- Extract geometric contour metrics: area (px and mm^2), perimeter, asymmetry index, compactness.
- Render 8 interactive visual diagnostic views (Views A through H).
- Conduct multi-turn text and voice consultations strictly grounded in 34 verified clinical practice guidelines.
- Search an audited directory of verified tertiary dermatology hospital departments in India.
- **What it CANNOT do:**
- It **cannot** diagnose cancer autonomously. Histopathologic examination remains the definitive gold standard.
- It **cannot** prescribe prescription pharmaceuticals (e.g., fluorouracil, imiquimod, pembrolizumab).
- It **cannot** evaluate non-dermoscopic, uncalibrated macroscopic smartphone images with high confidence.
- It **cannot** fabricate live hospital booking slots without real-world administrative coordination.

1.6 Memorizable Spoken Pitches

30-Second Elevator Pitch

> *"Professor, OptiDerma is an end-to-end multimodal dermatology AI decision-support platform. It takes a dermoscopic lesion image, validates its optical quality, digitally removes occluding hairs using DullRazor, and runs an EfficientNet-B4 neural network to classify seven lesion categories while segmenting the border. It provides visual explainability through Grad-CAM heatmaps, grounds patient consultations in 34 peer-reviewed guidelines via RAG, and coordinates verified hospital referrals—ensuring clinical explainability and patient safety at every step."*

One-Minute Presentation Pitch

> *"Professor, skin cancer screening currently faces two major problems: long specialist wait times that delay melanoma detection, and 'black-box' AI tools that output bare probabilities without explanation. OptiDerma solves both issues through a multimodal, clinical-first architecture.*
>
> *First, our computer vision pipeline combines an EfficientNet-B4 classifier with a U-Net segmentation decoder. Before inference, our custom 96-kernel DullRazor preprocessor removes hair artifacts to prevent spurious false positives. In addition to 7-class probabilities, the system extracts quantitative morphometrics—including border asymmetry and compactness.*

> *Second, we bridge the gap between vision and consultation using an Agentic RAG orchestrator. When a user asks about their lesion, the agent retrieves evidence from 34 indexed clinical guidelines from JAAD, BJD, and AAD using hybrid BM25 search. The consultation response is structured into six clinical segments with traceable citations, backed by a deterministic fallback that eliminates hallucinations. All sessions persist across restarts through a resilient dual-store SQLite and Supabase architecture."*

Three-Minute Comprehensive Academic Summary

> *"Good morning, Professor. I am presenting OptiDerma, a multimodal decision-support platform designed to assist healthcare professionals in cutaneous lesion evaluation.*
>
> *The clinical motivation is straightforward: melanoma has a 99% cure rate when caught early, but late detection is often fatal. Primary care doctors lack specialized dermoscopy training, leading to unnecessary biopsies or missed malignancies. Furthermore, existing AI classifiers suffer from artifact sensitivity and hallucinations.*
>
> *OptiDerma addresses these challenges through five tightly integrated architectural tiers:*
>
> *Tier 1 is our Pre-flight Image Quality Audit and DullRazor preprocessor. The system calculates Laplacian variance to reject blurred captures and uses 96 multi-scale directional morphological line kernels to isolate and inpaint hair strands, preserving genuine pigment networks.*
>
> *Tier 2 is our Multi-Task Deep Neural Network. We reverse-engineered and fine-tuned an EfficientNet-B4 backbone connected to both a 7-class classification head and a 4-stage convolutional transpose U-Net decoder. It simultaneously computes differential probabilities across the HAM10000 benchmark classes and extracts mathematical lesion geometry, including an asymmetry index based on principal-axis eigen-decomposition.*
>
> *Tier 3 is Explainable AI. We implemented analytical Grad-CAM from the final 448-channel feature map. Crucially, we do not just display a heatmap; we calculate a quantitative Spatial Energy Ratio, verifying that over 75% of saliency energy is concentrated within the segmented lesion contour.*
>
> *Tier 4 is our 10-Tool Agentic RAG Orchestrator. When answering clinical questions, the agent dynamically routes queries through hybrid BM25 and semantic keyword scoring across 34 peer-reviewed guidelines. It formats answers into six standardized clinical segments with verified citation badges, supported by an emergency red-flag triage filter and an offline deterministic fallback.*
>
> *Tier 5 is our Resilient Full-Stack Architecture: a Next.js 15 App Router frontend communicating with an asynchronous FastAPI backend, supporting both ElevenLabs WebRTC and server-side Whisper-to-TTS voice turns, backed by SQLite with Write-Ahead Logging and Supabase Cloud PostgreSQL.*
>
> *Our test suite has achieved 67 out of 67 passing backend tests, 0 TypeScript errors, and complete live verification across all nine primary pages. OptiDerma proves that AI in medicine must be explainable, grounded in verified literature, and designed to support—not replace—the clinician."*

Chapter 2: Problem Statement and Motivation

2.1 The Clinical Challenge of Cutaneous Lesion Assessment

The human skin is the body's largest organ, and differentiating between benign melanocytic proliferations (common nevi, seborrheic keratoses) and life-threatening malignancies (malignant melanoma, invasive basal cell carcinoma) is one of the most challenging visual tasks in clinical medicine.

Clinicians rely on the **ABCDE criteria** (Asymmetry, Border irregularity, Color variegation, Diameter $>6\text{mm}$, Evolution) and dermoscopic pattern analysis. However, visual evaluation by unaided eyes achieves diagnostic sensitivities of only **65% to 75%**. While epiluminescence dermoscopy increases sensitivity to $>85\%$, it requires extensive fellowship training. In rural and community clinics, non-specialist general practitioners often adopt a defensive biopsy approach, generating high healthcare costs and patient anxiety.

2.2 Why a Standalone Classifier is Insufficient

In academic computer vision literature, dozens of papers report $>90\%$ accuracy on benchmark datasets. However, a pure classification model is **clinically unusable** in real-world hospitals for several fundamental reasons:

1. **The Overconfidence Pathology:** As proven by Guo et al. (2017), modern deep neural networks with batch normalization and high capacity are miscalibrated. An uncalibrated softmax score of 0.99 simply means the logit margin was wide, not that the patient has a 99% probability of disease.

2. **Spurious Correlation & Shortcut Learning:** Convolutional networks frequently learn non-biological artifacts present in training datasets, such as surgical skin markings, dermatoscope contact ring shadows, gel bubbles, and overlying hair fibers. If a model predicts "melanoma" because it detected hair intersections mimicking a pigment network, clinical reliance on it is hazardous.

3. **The "Actionability" Void:** A doctor or patient who receives the output `mel: 0.88` is left stranded. The model does not specify *what anatomical features* triggered the alert, *what surgical excision margins* are recommended by clinical guidelines, or *which certified local hospital* can perform the biopsy.

2.3 The Necessity of Explainability, Literature RAG, and Verified Referral

To transform an academic classifier into a clinically viable decision-support system, four complementary technologies must be unified:

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■ Diagnostic Vision ■ Explainable AI (XAI) ■ Guideline RAG ■
■ EfficientNet-B4 7-cls ■ Grad-CAM + Spatial Energy ■ 34 JAAD/BJD Guidelines■
■ U-Net Segmentation ■ DullRazor 96-kernel Filter ■ Zero Hallucination ■
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1. **Optical Pre-Screening & Hair Removal (DullRazor):** Removing obscuring artifacts prior to feature extraction guarantees that the deep feature representations represent dermal tissue rather than hair keratin.

2. **Mathematical Explainability & Morphometrics:** Quantifying border asymmetry and checking that Grad-CAM saliency energy is localized inside the lesion boundary provides verifiable evidence to the clinician.

3. **Retrieval-Augmented Generation (RAG):** Bridging visual findings with peer-reviewed literature gives the practitioner immediate access to evidence-based management protocols (e.g., Breslow thickness guidelines, 2mm excisional biopsy recommendations).

4. **Care Connect:** Channeling the user toward verified tertiary healthcare institutions closes the loop between automated assessment and actual clinical care.

2.4 Distinguishing Motivation from Demonstrated Clinical Proof

It is essential to maintain academic integrity:

- **Project Motivation:** To explore whether combining deep vision, morphological XAI, and literature RAG can improve the transparency and triage efficiency of skin cancer evaluation.

- **Clinical Reality:** OptiDerma is an experimental engineering research platform. It has been validated through rigorous automated software tests (67/67 unit and integration tests), mathematical XAI sanity checks, and benchmark dataset inference. It has **not** undergone prospective multi-center randomized clinical trials (RCTs) or FDA/CE regulatory clearance. All outputs represent decision support, not medical diagnosis.

Chapter 3: Complete System Architecture

3.1 Comprehensive End-to-End System Architecture Diagram

```
flowchart TD
subgraph CLIENT["Frontend Client Tier (Next.js 15 App Router)"]
UI_DASH["Interactive Dashboard\n(Three.js Brain Canvas)"]
UI_UP["Upload Zone\n(Pre-flight IQA Audit)"]
UI_RES["Results View\n(7-Class Probs & Risk Badge)"]
UI_CANVAS["Lesion Canvas (Views A-H)\n(Grad-CAM, Morphometrics, DullRazor)"]
UI_CHAT["6-Segment AI Consultation\n(Mode A: General / Mode B: Analysis-Aware)"]
UI_HIST["Consultation History Drawer\n(Session Resumption & Search)"]
UI_VOICE["Voice Assistant Console\n(ElevenLabs WebRTC & Voice-Turn)"]
UI_CARE["Care Connect Modal\n(Verified Clinic Directory Search)"]
end

subgraph GATEWAY["API Gateway & Security (FastAPI 0.115)"]
SEC["Security & Rate-Limit Middleware\n(20MB Cap, 300 req/min, CSP, nosniff)"]
ROUTER["Unified API Router (/api/v1 & Direct)"]
end

subgraph VISION["Computer Vision & Explainable AI Tier"]
IQA["Image Quality Assessment\n(Laplacian Blur, Brightness, Glare)"]
DULL["DullRazor Preprocessor\n(96 Directional Kernels + Telea Inpaint)"]
PYTORCH["PyTorch 2.6 OptiDermaNet\n(EfficientNet-B4 + U-Net, 17.7M Params)"]
GRADCAM["Grad-CAM Saliency Engine\n(Autograd on f4 448ch + Bilinear Upscale)"]
SPATIAL["Spatial XAI Energy Diagnostics\n(Energy Inside %, Spatial Ratio)"]
MORPH["Morphometric Geometry Engine\n(Eigen-Asymmetry, Compactness, Area mm²)"]
VERIF["Independent Verification Audit\n(Deterministic Checksum & Bounds Check)"]
end

subgraph AGENT_TIER["Knowledge & Agentic AI Orchestrator"]
ORCH["Central Agentic Orchestrator\n(optiderma_workflow, 10 Typed Tools)"]
ROUTE_INTENT["Dynamic Intent Classifier\n(Emergency, Care, Report, Analysis QA)"]
RAG["Dermatology RAG Service\n(34 Guidelines, BM25 + Semantic Boost)"]
SAFETY["Emergency Safety Filter\n(Red-flag triage, Bleeding/Ulceration)"]
LLM["Groq Llama-3.3-70B\n(Temperature 0.2, Max 3500 Tokens)"]
FALLBACK["Deterministic Clinical Fallback\n(Zero-Hallucination Truthful Synthesis)"]
end

subgraph AUDIO["Multimodal Audio Tier"]
STT["Whisper Speech-to-Text\n(Audio Transcription)"]
CLEAN_TXT["Speech Sanitizer\n(Stripping Markdown & Citations)"]
TTS["ElevenLabs / gTTS Engine\n(Streaming MP3 Audio Generation)"]
end

subgraph PERSIST["Dual-Tier Persistence Tier"]
SQLITE["Local SQLite Store\n(backend/data/optiderma_persistence.db, WAL)"]
SUPABASE["Supabase Cloud PostgreSQL\n(PostgREST REST Client, Dual-Write Sync)"]
end

UI_UP -->|Multipart JPEG/PNG/WEBP| SEC
UI_CHAT -->|JSON Conversation Payload| SEC
UI_VOICE -->|Audio Blob / WebRTC| SEC
UI_CARE -->|Search / Booking Form| SEC

SEC --> ROUTER
ROUTER -->|/predict & /analyze| IQA
IQA --> DULL
DULL --> PYTORCH
PYTORCH --> GRADCAM
PYTORCH --> MORPH
GRADCAM --> SPATIAL
SPATIAL --> VERIF

ROUTER -->|/conversation & /agent/voice-turn| ORCH
ORCH --> SAFETY
ORCH --> ROUTE_INTENT
ROUTE_INTENT --> RAG
ROUTE_INTENT --> LLM
LLM -.->|Connection Failure| FALLBACK
RAG --> ORCH
```

```
ROUTER -->|/agent/voice-turn| STT
STT --> ORCH
ORCH --> CLEAN_TXT
CLEAN_TXT --> TTS
TTS -->|MP3 Audio Stream| UI_VOICE

VERIF --> PERSIST
ORCH --> PERSIST
PERSIST --> SQLITE
PERSIST --> SUPABASE
```

3.2 Detailed Component Architectural Audit (The 7-Question Breakdown)

1. Next.js 15 App Router Frontend

- **What it is:** High-performance React 19 web application built with TypeScript, TailwindCSS, and Lucide icons.
- **Why it is needed:** Provides an interactive, dark-mode clinical workstation interface for physicians and patients.
- **Input it receives:** User mouse/touch interactions, uploaded dermoscopic image files, text queries, microphone audio streams, and filter parameters.
- **Processing it performs:** Pre-inference client-side image inspection, state management for active analyses, rendering of 8 lesion canvas views, dynamic markdown rendering, and slide-out drawer management.
- **Output it produces:** Rendered HTML5/DOM, multipart HTTP requests, and WebSocket/WebRTC audio streams.
- **File implementing it:** `app/page.tsx`, `components/opti-derma-dashboard.tsx`, `components/lesion-canvas.tsx`, `components/consultation-chat.tsx`.
- **Component calling it:** Directly accessed by the user's web browser (`http://localhost:3000`).

2. FastAPI Backend & Gateway

- **What it is:** Asynchronous Python 3.12 REST API server running on Uvicorn.
- **Why it is needed:** Acts as the central integration hub, executing deep neural inference, orchestrating RAG queries, and enforcing security policies.
- **Input it receives:** HTTP GET, POST, PATCH, and DELETE requests from the Next.js client.
- **Processing it performs:** IP rate limiting (300 requests/minute), 20MB payload enforcement, security header injection, and dispatching to vision and agent services.
- **Output it produces:** JSON responses adhering to Pydantic schemas, streaming MP3 audio, and downloadable PDF reports.
- **File implementing it:** `backend/app/main.py`, `backend/app/api/router.py`.
- **Component calling it:** Next.js API client (`lib/api.ts`).

3. Image Quality Assessment (IQA) & DullRazor

- **What it is:** Computer vision preprocessing module combining OpenCV analytical algorithms and morphological hair filters.
- **Why it is needed:** Rejects unreadable images before wasting GPU/CPU cycles and eliminates occluding hair strands that trigger false-positive predictions.
- **Input it receives:** Raw decoded BGR image matrix (`np.ndarray`).
- **Processing it performs:** Laplacian variance blur estimation ($\sigma^2_{\text{Laplacian}} \ge 12.0$), specular glare masking (>248 brightness), 96-directional morphological black-hat hair filtering, and Telea fast-marching inpainting.
- **Output it produces:** `QualityReport`, `clean_bgr` image, `hair_mask`, strand counts, and PSNR diagnostics.
- **File implementing it:** `model/preprocessor.py`.
- **Component calling it:** `backend/app/services/inference_service.py:run_prediction()`.

4. PyTorch EfficientNet-B4 + U-Net Inference Engine

- **What it is:** Multi-task deep convolutional neural network with 17,740,464 parameters.
- **Why it is needed:** Performs 7-class lesion probability classification and generates a pixel-level lesion segmentation mask.
- **Input it receives:** Standardized float32 tensor of shape `(1, 3, 256, 256)`, normalized using ImageNet mean `[0.485, 0.456, 0.406]` and std `[0.229, 0.224, 0.225]`.
- **Processing it performs:** Compound-scaled feature extraction across 5 convolutional stages, global pooling with linear classification, and 4-stage convolutional transpose decoding with skip connections.
- **Output it produces:** Raw class logits `(1, 7)`, binary segmentation mask logits `(1, 1, 256, 256)`, and stage-5 feature maps `f4` `(1, 448, 8, 8)`.
- **File implementing it:** `model/model_pipeline.py:OptiDermaNet`, `model/model_pipeline.py:OptiDermaPipeline`.
- **Component calling it:** `backend/app/services/inference_service.py:run_prediction()`.

5. Explainable AI (Grad-CAM) & Morphometrics Engine

- **What it is:** Analytical gradient backpropagation module and geometric contour analyzer.
- **Why it is needed:** Explains *where* the model focused and computes physical lesion shape metrics.
- **Input it receives:** Feature tensor `f4`, class logit y^c, and binary segmentation mask.
- **Processing it performs:** Autograd gradient computation, global average pooling to weights w_k^c, ReLU activation, bilinear upsampling to 256×256, Spatial Energy Ratio calculation, and principal-axis eigen-decomposition for asymmetry.
- **Output it produces:** Normalized 256×256 float32 CAM heatmap, Asymmetry Index ($0.0 \dots 1.0$), Compactness ($P^2 / 4\pi A$), Area, Perimeter, and 8 visual Data URL views.
- **File implementing it:** `model/model_pipeline.py:compute_cam()`, `model/model_pipeline.py:compute_metrics()`.
- **Component calling it:** `backend/app/services/inference_service.py`.

6. Central Agentic Orchestrator & Dermatology RAG

- **What it is:** Stateful clinical dialogue engine managing 10 typed Pydantic tools, 5 dynamic routes, and 34 indexed guidelines.
- **Why it is needed:** Synthesizes evidence-grounded consultations without hallucinating medical facts or fabricating citations.
- **Input it receives:** User query text, active `conversation_id`, active `analysis_id`, and conversation history.
- **Processing it performs:** Red-flag emergency screening, intent classification, hybrid Robertson-Spärck Jones BM25 retrieval, condition code boosting (+2.0), prompt assembly, Groq LLM inference, and deterministic fallback execution.
- **Output it produces:** 6-segment structured clinical response, citation objects `[CIT-X]`, tool telemetry records, and updated conversation history.
- **File implementing it:** `backend/app/agents/optiderma_agent.py`, `backend/app/rag/dermatology_rag.py`.
- **Component calling it:** `backend/app/main.py:handle_conversation_turn()`.

7. Dual-Tier Persistence Service

- **What it is:** Repository-pattern database service combining local SQLite with remote Supabase Cloud PostgreSQL.
- **Why it is needed:** Ensures that every analysis record, visual artifact, conversation message, and appointment referral survives server restarts with zero data loss, even during complete offline operation.
- **Input it receives:** Analysis dictionaries, artifact base64 strings, conversation messages, and booking schemas.
- **Processing it performs:** UUID canonicalization, WAL-mode SQLite execution, thread-safe memory caching, and PostgREST HTTPS dual-writing.
- **Output it produces:** Persisted database records, conversation history arrays, and appointment referral records.
- **File implementing it:** `backend/app/services/persistence_service.py`.
- **Component calling it:** `backend/app/services/inference_service.py`, `backend/app/agents/optiderma_agent.py`.

3.3 Complete Request / Data Flow

Flow 1: Complete Image-Based Analysis Request

```
[User Browser]
■ 1. Drag & drop lesion image; click "Run Model Inference"
▼
[FastAPI /predict]
■ 2. SecurityMiddleware checks Content-Length (<20MB) and IP rate limits
▼
[Inference Service]
■ 3. cv2.imdecode parses bytes into BGR uint8 matrix
■ 4. preprocessor.assess_quality evaluates Laplacian blur, glare, and resolution
■ 5. preprocessor.remove_hair executes 96-kernel DullRazor and Telea inpainting
■ 6. Tensor normalized: (img - mean) / std -> (1, 3, 256, 256)
▼
[PyTorch OptiDermaNet]
■ 7. Forward pass -> cls_logits (1, 7), seg_logits (1, 1, 256, 256), f4 (1, 448, 8, 8)
■ 8. Softmax with temperature scaling -> top-3 differential probabilities
■ 9. autograd.grad(cls_logits[0, target_idx], f4) -> Grad-CAM heatmap (256x256)
■ 10. Sigmoid on seg_logits -> binary mask -> compute_metrics (Asymmetry, Area, Compactness)
■ 11. Spatial XAI Energy calculation: Energy Inside %, Spatial Ratio
▼
[Verification Service]
■ 12. Checks SHA-256 checkpoint hash (006a77b02a) and probability bounds
▼
[Persistence Service]
■ 13. Writes analysis record and 8 visual Data URLs to SQLite WAL & Supabase
▼
[Client Response]
■ 14. Next.js receives JSON -> renders probabilities, risk badge, and Lesion Canvas
```

Flow 2: Multi-Turn Evidence-Grounded Consultation Request

```
[User Browser]
■ 1. Types: "Why is this lesion high risk and what do guidelines say?"
▼
[FastAPI /conversation]
■ 2. Dispatches to optiderma_workflow.run_turn()
▼
[Agent Orchestrator]
■ 3. Tool 4 (evaluate_safety_and_urgency) scans for bleeding or rapid growth
■ 4. Classifies intent: route = 'analysis_qa' (analysis_id present)
■ 5. Retrieves canonical analysis record from PersistenceService
■ 6. Tool 2 (retrieve_dermatology_evidence):
■ - Tokenizes query; strips stop words
■ - Runs BM25 scoring over 34 guidelines
■ - Applies condition boost for predicted class (e.g., 'mel')
■ - Returns top-2 evidence passages with verified IDs
■ 7. Tool 3 (generate_evidence_grounded_answer):
■ - Injects system prompt with strict 6-segment format
■ - Calls Groq API (llama-3.3-70b-versatile)
■ - On API failure -> triggers deterministic literature synthesis fallback
■ 8. Tool 5 (save_conversation_turn):
■ - Persists user prompt and assistant response with [CIT-1] citations to SQLite
▼
[Client Response]
■ 9. UI renders 6-segment response and interactive citation drawer
```

Chapter 4: Frontend — Page-by-Page Teacher Demonstration Guide

This chapter provides the exact manual for demonstrating every major page and feature of OptiDerma during your viva or presentation tomorrow.

Master Navigation & Route Reference Table

Step	Visible Feature	Actual URL / Route	Demonstration Action	Expected Result	Technical Implementation
1	**Main Platform Dashboard**	`http://localhost:3000/`	Open homepage, show 3D brain, point to system status badges	Dark-mode clinical UI, live status badges (`Model: Ready`, `RAG: 34 Guidelines`)	Next.js 15 App Router, Three.js canvas, TailwindCSS
2	**Image Upload & IQA Audit**	`http://localhost:3000/` (scrolls to `#upload`)	Select sample lesion file; review IQA metrics; click "Run Model Inference"	Upload preview renders with dimensions and sharpness; button triggers API	`components/upload-zone.tsx`, `model/preprocessor.py`
3	**Analysis & Results View**	`http://localhost:3000/` (renders `#results`)	Review differential probabilities, risk badge, and verification audit	7 horizontal probability bars, High Risk badge, verification checkmark	`components/results-view.tsx`, PyTorch EfficientNet-B4
4	**Explainability Lesion Canvas**	`http://localhost:3000/` (canvas below results)	Toggle between Views A–H; zoom/pan; inspect morphometrics panel	Renders DullRazor clean image, Grad-CAM overlay, Asymmetry Index ($0.0\dots1.0$)	`components/lesion-canvas.tsx`, OpenCV morphology, autograd
5	**Text Consultation & RAG**	`http://localhost:3000/consultation`	Type clinical question; click send; click citation badge `[CIT-1]`	Structured 6-segment response rendered; citation drawer expands	`components/consultation-chat.tsx`, Agentic RAG, BM25
6	**Session History Drawer**	`http://localhost:3000/` (Click "History" icon)	Click clock icon; view saved sessions; resume prior conversation	History drawer slides out; prior messages and analysis context restored	`components/consultation-history-drawer.tsx`, SQLite WAL
7	**Voice Assistant Console**	`http://localhost:3000/` (Click "Voice Assistant")	Open voice modal; inspect visualizer ring and transcription state	Voice modal displays readiness; supports WebRTC and spoken voice-turn	`components/voice-assistant.tsx`, Whisper STT, ElevenLabs
8	**Care Connect Directory**	`http://localhost:3000/` (Click "Care Connect")	Enter PIN `110029` or select Delhi; view AIIMS; click "Request Consultation"	Verified clinic cards render; appointment referral form opens	`components/care-connect-modal.tsx`, `care_service.py`

Detailed Walkthrough for Each Page

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Page 1: Main Platform Dashboard ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

- **A. What the teacher sees:**

A modern, high-contrast dark clinical dashboard. At the center is an interactive 3D brain and neural network canvas. At the top right are three glowing operational status badges: `Model: Ready (EfficientNet-B4)`, `RAG: 34 Guidelines`, and `Dual-Store: Active`. Four interactive cards invite action: *New Skin Analysis*, *AI Dermatology Consultation*, *Voice Assistant*, and *Care Connect*.

- **B. What you should say:**

> *"Professor, this is the main clinical dashboard of OptiDerma. At a glance, the clinician can verify system integrity through our live backend status badges: our multi-task PyTorch model is pre-warmed in memory, our 34 clinical practice guidelines are indexed for retrieval, and our dual-store persistence engine is operational. Let us start by analyzing a suspicious dermoscopic lesion."*

- **C. What you should click:**

1. Hover over the operational status badges to demonstrate live backend connectivity.

2. Click the **"New Skin Analysis"** card or button.

- **D. What happens behind the scenes:**

The browser executes an asynchronous heartbeat to `GET http://127.0.0.1:8000/health` and `GET http://127.0.0.1:8000/ready`. The frontend receives JSON confirming that checkpoint weights (`006a77b02a`) are loaded and transitions the UI to the upload workstation.

- **E. What result to expect:**

Smooth animated scroll down to the `#upload` zone with all diagnostic controls active.

- **F. What makes it technically interesting:**

The dashboard integrates WebGL / Three.js for anatomical visualization without degrading page performance, utilizing Next.js 15 Turbopack for sub-second hot reloading.

- **G. Likely teacher questions & answers:**

1. *Q: Why is there a 3D visualization on a dermatology dashboard?*

A: It serves as an anatomical orientation model for future body-mapping teledermatology extensions, helping clinicians localize lesions on specific cutaneous regions.

2. *Q: How does the dashboard know the backend is ready?*

A: It queries `GET /ready`*, which performs an in-memory check verifying that the PyTorch state dictionary is resident in RAM.*

3. *Q: Is this responsive on mobile tablets used by hospital nurses?*

A: Yes, the interface uses fluid TailwindCSS grid breakpoints (`grid-cols-1 md:grid-cols-2 lg:grid-cols-4`*) optimized for clinical touchscreens.*

- **H. Failure recovery:**

If the status badge shows `Backend Offline`, open your terminal and verify that `python -m uvicorn app.main:app --host 127.0.0.1 --port 8000` is running.

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Page 2: Image Upload & Pre-Flight IQA ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

- **A. What the teacher sees:**

A targeted drag-and-drop upload zone with dashed teal borders. Directly beneath are four Image Quality Audit (IQA) threshold badges: *Resolution ($\ge 150\text{px}$)*, *Sharpness (Laplacian $>12$)*, *Glare ($<12\%$)*, and *Contrast*. A sample image preset button (`test_lesion.jpg`) is visible.

- **B. What you should say:**

> *"A critical problem with clinical AI is 'garbage-in, garbage-out'—if a nurse uploads an out-of-focus or glare-saturated photograph, deep models can make catastrophic errors. OptiDerma enforces a fail-fast Image Quality Audit before allowing inference. Let us load a sample high-resolution dermoscopic image."*

- **C. What you should click:**

1. Click **"Choose File"** and select `backend/tests/fixtures/sample_melanoma.jpg` (or click the sample lesion button).

2. Point to the image preview showing detected resolution ($1024 \times 768$) and calculated sharpness score.

3. Click the bright teal button: **"Run Model Inference"**.

- **D. What happens behind the scenes:**

The Next.js client packs the raw uncompressed image into a `FormData` multipart payload and streams it via `POST /predict`. In the backend, OpenCV parses the buffer, executes `preprocessor.assess_quality()`, runs the 96-kernel DullRazor filter, and normalizes the tensor into PyTorch.

- **E. What result to expect:**

A progress spinner displays *"Analyzing Lesion Morphometry & Computing Attributions..."* for approximately 2 to 3.5 seconds on CPU, followed by smooth rendering of the results section.

- **F. What makes it technically interesting:**

The IQA pipeline uses mathematical Laplacian convolution (`cv2.Laplacian`) to compute edge variance, instantly rejecting blurred images without expensive deep learning computation.

- **G. Likely teacher questions & answers:**

1. *Q: What happens if I upload a blurred selfie?*

A: The Laplacian variance drops below 12.0. The backend halts inference with HTTP 422 and returns actionable advice: 'Image is severely blurred; please increase illumination and capture a focused dermoscopic image.'

2. *Q: What image formats are supported?*

A: Standard dermoscopic raster formats: JPEG, PNG, and WEBP up to a 20MB limit.

3. *Q: Can a user bypass the quality audit if necessary?*

A: Yes, clinical administrators can pass `?force=true` to force inference on difficult borderline images.

- **H. Failure recovery:**

If the upload fails with a 413 error, explain to the teacher that FastAPI's `SecurityAndRateLimitMiddleware` strictly enforces a 20MB ceiling to prevent buffer-overflow and denial-of-service attacks.

```
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■ Page 3: Analysis & Diagnostic Results ■
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```

- **A. What the teacher sees:**

The diagnostic results dashboard (`.results-view`). At the top is a large **Malignancy Risk Stratification Card** with a glowing red indicator: `High Risk — Immediate Clinical Biopsy & Urgent Dermatologic Consult Recommended`. Below is a horizontal bar chart showing the **Top Differential Diagnoses** across all seven HAM10000 classes:

- `Cutaneous Melanoma (mel): 92.4%`
- `Basal Cell Carcinoma (bcc): 4.1%`
- `Benign Keratosis-like Lesions (bkl): 1.8%`
- `Melanocytic Nevi (nv): 0.9%`

A verification badge shows: `Verification: Passed (Checksum 006a77b02a)`.

- **B. What you should say:**

> *"Here, the teacher can see the raw model outputs converted into clinical decision support. Notice that we do not simply display 'Cancer: Yes'. The system provides a full 7-class differential diagnosis with probabilities. Furthermore, our risk stratification engine computes the normalized Shannon entropy across the top-3 predictions. Because melanoma exceeds 30% confidence, it is automatically categorized as High Risk."*

- **C. What you should click:**

1. Point to the primary prediction bar (`mel`).

2. Click **"View Verification Audit Details"** to show the cryptographic hash (`006a77b02a`) and image SHA-256 fingerprint.

3. Point to the action buttons: *"Export PDF Report"* and *"Open AI Consultation"*.

- **D. What happens behind the scenes:**

The backend executes `stratify_risk()` based on class rules and entropy:

$$\text{Entropy} = -\sum \frac{p_i \ln p_i}{\ln 3}$$

It calls `verification_service.verify_analysis()`, verifying that probabilities sum to $1.0 \pm 0.001$, no NaN values exist, and the checkpoint matches the authorized hospital hash.

- **E. What result to expect:**

Accurate, stabilized differential diagnosis probabilities with an independent verification checkmark.

- **F. What makes it technically interesting:**

The dual evaluation: standard cross-entropy classification is paired with an independent deterministic verification audit, satisfying FDA Software as a Medical Device (SaMD) traceability principles.

- **G. Likely teacher questions & answers:**

1. *Q: Why is melanoma considered High Risk at only 30% confidence?*

A: Because of asymmetric clinical loss. Missing a melanoma (false negative) is potentially fatal, whereas performing a biopsy on a benign mole (false positive) carries minimal morbidity. Clinical utility demands a highly sensitive threshold for melanoma.

2. *Q: Where do the 7 classes come from?*

A: They are the exact diagnostic classes from the international HAM10000 benchmark dataset.

3. *Q: What is the purpose of the verification audit?*

A: It guarantees that the inference output was generated by the approved model checkpoint without in-flight tensor corruption or bit-flip errors.

- **H. Failure recovery:**

If the results do not render, check browser console `F12` for network errors. If `fetch /predict` failed, confirm the backend process is active in terminal 1.

```
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■ Page 4: Explainability (XAI) & Lesion Canvas ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

- **A. What the teacher sees:**

The high-resolution interactive **Lesion Canvas** with 8 selectable view tabs:

- *View A: Original Dermoscopy*
- *View B: Raw Grad-CAM Activation Heatmap*
- *View C: Overlay Attention Map*
- *View D: Lesion Boundary & U-Net Contour*
- *View E: Saliency Comparison*
- *View F: Multi-Class Attributions*
- *View G: DullRazor Cleaned Field*
- *View H: Occluding Hair Diagnostic Mask*

To the right is the **Morphometrics Telemetry Panel**, displaying:

- *Area:* $14,250\text{ px} \ (48.8\text{ mm}^2)$
- *Perimeter:* $482\text{ px} \ (28.2\text{ mm})$
- *Asymmetry Index:* $0.42$ (Moderate to High asymmetry)
- *Compactness:* $2.84$ (Marked border irregularity)
- *Spatial Energy Ratio:* $4.8\times$ ($78.4\%$ energy localized inside lesion)

- **B. What you should say:**

> *"This is one of OptiDerma's core scientific contributions: opening the neural network's black box. First, let us inspect View G: DullRazor. Notice how the preprocessor detected overlying hair fibers, created a diagnostic mask in View H, and inpainted the clean skin texture. Next, look at View C: Overlay. Our Grad-CAM backpropagates gradients from the 448-channel convolutional head to show exactly where the model focused. Most importantly, our Spatial Energy Ratio proves that 78.4% of the attribution energy is concentrated inside the lesion boundary—confirming the network is not relying on background lighting or border artifacts."*

- **C. What you should click:**

1. Click **"View G: DullRazor"** to demonstrate the hair-inpainted clean field.
2. Click **"View H: Hair Mask"** to show the segmented hair strands.
3. Click **"View C: Overlay"** to show the JET heatmap blended over the lesion.
4. Use the zoom slider (`+` / `-`) or mouse wheel to zoom into the pigment network.

- **D. What happens behind the scenes:**

The canvas dynamically swaps between the base64 Data URLs generated during `run_prediction()`. For Grad-CAM, PyTorch’s autograd calculated $\frac{\partial y^{\text{mel}}}{\partial f4}$, pooled channel weights, and performed bilinear interpolation to $256 \times 256$. Morphometric contour coordinates are drawn as an SVG path.

- **E. What result to expect:**

Instant, flicker-free rendering of the selected visual diagnostic view with interactive zoom and pan controls.

- **F. What makes it technically interesting:**

The mathematical synergy between the U-Net segmentation mask and the Grad-CAM saliency map: instead of merely displaying a qualitative heatmap, we calculate the exact integral of saliency energy within the segmented polygon.

- **G. Likely teacher questions & answers:**

1. *Q: How is the Asymmetry Index calculated?*

A: We extract the lesion contour, compute the principal inertia axes using eigen-decomposition of the spatial covariance matrix, rotate the mask, flip it along both axes, and calculate the XOR intersection over union.

2. *Q: What does a Compactness score of 2.84 mean?*

A: Compactness is defined as $P^2 / (4\pi A)$. For a perfect circle, the value is 1.0. A score of 2.84 indicates significant border irregularity, a classic ABCDE criterion for cutaneous melanoma.

3. *Q: What if Grad-CAM focuses outside the lesion?*

A: If the energy inside is $<40\%$, the system flags an 'Attribution Bleed' warning, alerting the clinician that the prediction may be unreliable.

- **H. Failure recovery:**

If canvas views appear blank, verify that the backend returned the `images` dictionary with base64 Data URLs.

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Page 5: Multi-Turn AI Consultation & RAG ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

- **A. What the teacher sees:**

The AI Dermatology Consultation interface (`/consultation`). The chat is operating in **Mode B: Analysis-Aware Assistant**, displaying a pill badge: `Active Context: Analysis #uuid`. The chat contains a rich 6-segment structured clinical response with bold headings and blue bracketed citation badges (`[CIT-1]`, `[CIT-2]`). At the bottom is an input box with suggested follow-up prompts.

- **B. What you should say:**

> *"Now we move to our conversational intelligence layer. Unlike general-purpose chatbots that produce unstructured text and invent medical claims, OptiDerma enforces a strict 6-segment clinical response protocol. It grounds every statement in our 34 indexed peer-reviewed dermatology guidelines. Notice that clicking citation badge [CIT-1] expands the exact publication from the American Academy of Dermatology."*

- **C. What you should click:**

1. In the chat box, type: `"Explain the clinical management guidelines for this lesion."`

2. Click the send icon.

3. When the assistant responds, click on the `[CIT-1]` badge to open the citation drawer.

- **D. What happens behind the scenes:**

The request hits `POST /conversation`. The orchestrator executes `evaluate_safety_and_urgency`, runs BM25 retrieval over the 34 guidelines, applies a condition boost for `mel`, injects the retrieved passages into Groq Llama-3.3-70B, and streams the 6-segment answer back. If Groq times out, our deterministic literature synthesis executes immediately.

- **E. What result to expect:**

A response structured into:

- *Segment 1: Executive Diagnostic Correlation*
- *Segment 2: Dermoscopic Feature Attribution*
- *Segment 3: Evidence-Based Clinical Guidance* `[CIT-1]`
- *Segment 4: Patient-Specific Risk Stratification*

- *Segment 5: Questions for In-Person Dermatologist*
- *Segment 6: Statutory Research Disclaimer & Action Plan*
- **F. What makes it technically interesting:**

The deterministic fallback engine: even with zero internet or API keys, the system synthesizes 100% factual guidance from local literature guidelines without hallucination.

- **G. Likely teacher questions & answers:**

1. *Q: How do you prevent the LLM from making up fake medical citations?*

A: The prompt strictly confines the LLM to supplied context passages with immutable IDs. In fallback mode, citations are mapped programmatically from our indexed repository.

2. *Q: What is the difference between Mode A and Mode B?*

A: Mode A is general dermatology education without an active image. Mode B is analysis-aware, automatically injecting the active lesion's morphometrics, predicted class, and asymmetry index.

3. *Q: What happens if a patient asks how to cut the mole off at home?*

A: The system’s safety filter triggers an immediate refusal, reiterating that surgical excision requires sterile hospital conditions and narrow-margin histopathology.

- **H. Failure recovery:**

If the Groq API key is invalid or rate-limited, demonstrate the fallback by explaining: *"OptiDerma detected cloud API latency and seamlessly transitioned to local deterministic literature synthesis."*

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Page 6: Consultation History & Resumption ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

- **A. What the teacher sees:**

A slide-out drawer on the right side of the screen (`components/consultation-history-drawer.tsx`). It lists past consultation sessions with readable auto-generated titles (e.g., *"Melanoma Risk Analysis & Guidelines"*), timestamps, message counts, and linked analysis badges (`Analysis #uuid`). A search bar and delete controls are visible.

- **B. What you should say:**

> *"Clinical workflows require continuity of care. OptiDerma persists all consultation sessions, messages, and linked analyses in our dual-store repository. Even if the browser is closed or the server restarts, a doctor can reopen the history drawer and resume any past consultation with full diagnostic context intact."*

- **C. What you should click:**

1. Click the clock icon **"History"** in the top navigation bar.
2. Point out the list of saved sessions.
3. Click on a past session to demonstrate instant resumption.

- **D. What happens behind the scenes:**

The frontend calls `GET /conversations`. When a session is clicked, it calls `GET /conversations/{id}`. `PersistenceService` queries the SQLite database (`conversations` and `conversation_messages` tables), reconstructs the message thread, and re-links the associated `analysis_id`.

- **E. What result to expect:**

The chat interface updates instantly, restoring the full conversation history, citations, and active analysis context.

- **F. What makes it technically interesting:**

The foreign key relationship in SQLite and Supabase: `conversations.analysis_id` references `analyses.id`, ensuring conversations are never disconnected from their ground-truth computer vision telemetry.

- **G. Likely teacher questions & answers:**

1. *Q: Where is this data physically stored?*

A: In our local SQLite database at `backend/data/optiderma_persistence.db` *using Write-Ahead Logging, replicated to Supabase Cloud PostgreSQL.*

2. *Q: If I delete a conversation, does it delete the patient's analysis?*

A: No. We enforce referential integrity: deleting a conversation removes the message history, but the underlying clinical image analysis is permanently retained for medical auditability.

3. *Q: How do you prevent cross-conversation context leakage?*

A: Each session has an isolated UUID. When switching sessions, the frontend clears active memory state and re-initializes context exclusively from the selected session ID.

- **H. Failure recovery:**

If the drawer is empty, send a message in the chat to create a new session; it will appear in the drawer immediately.

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Page 7: Voice Assistant & Audio Turn ■
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```

- **A. What the teacher sees:**

The **Voice Assistant Console** modal. In the center is an animated audio visualizer ring. A mode selector allows toggling between *ElevenLabs Conversational WebRTC* and *Backend Spoken Voice-Turn Pipeline*. Live transcript boxes display speech input and agent response.

- **B. What you should say:**

> *"For accessibility and sterile clinical environments where doctors cannot touch a keyboard, OptiDerma provides hands-free voice interaction. We support direct ElevenLabs WebRTC streaming as well as our server-side spoken voice-turn pipeline, which transcribes speech via Whisper, executes agent reasoning with safety triage, and synthesizes natural speech via ElevenLabs or Google TTS."*

- **C. What you should click:**

1. Click **"Voice Assistant"** in the top navigation bar.

2. Open the console to show the visualizer ring and audio device controls.

3. Point out the live transcript pane.

- **D. What happens behind the scenes:**

When an audio file is uploaded to `POST /api/v1/agent/voice-turn`, the backend calls `WhisperSTTService` for transcription, routes text to `optiderma_workflow.run_turn()`, sanitizes Markdown syntax via `clean_text_for_speech()`, and streams an MP3 audio file generated by ElevenLabs or `gTTS`.

- **E. What result to expect:**

The voice assistant modal renders with live audio visualizer feedback and ready status.

- **F. What makes it technically interesting:**

The text-to-speech sanitization engine: raw clinical markdown containing bold tags, tables, and citation brackets (`[CIT-1]`) sounds robotic if read aloud. Our `clean_text_for_speech()` utility converts medical markdown into natural, spoken English dialogue before audio synthesis.

- **G. Likely teacher questions & answers:**

1. *Q: What happens if ElevenLabs is unavailable?*

A: The system automatically falls back to local Google Text-to-Speech (`gTTS`), streaming an MP3 buffer without crashing.

2. *Q: What speech-to-text model is used?*

A: OpenAI Whisper (`base` or `small` model), providing robust medical terminology transcription.

3. *Q: Can voice input trigger emergency triage?*

A: Yes. The transcribed text is evaluated by `evaluate_safety_and_urgency` before any response is generated.

- **H. Failure recovery:**

If microphone permissions are blocked in the browser, demonstrate the voice pipeline by pointing to our passing automated test suite: `pytest backend/tests/test_voice_turn_pipeline.py` (7/7 tests passed).

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Page 8: Care Connect & Clinician Referral ■
```

- **A. What the teacher sees:**

The **Care Connect** modal (`components/care-connect-modal.tsx`). It features a geographic search bar with filters for City, District, and PIN code. Verified hospital cards are displayed with official badges:

- *AIIMS New Delhi — Department of Dermatology, Venereology & Leprology* (Ansari Nagar, New Delhi)
- *PGIMER Chandigarh — Department of Dermatology* (Sector 12, Chandigarh)
- *KEM Hospital Mumbai — Dermatology & Dermatosurgery Clinic* (Parel, Mumbai)
- *Victoria Hospital / BMCRI Bengaluru — Dermatology OPD* (Fort Road, Bengaluru)

Each card displays verified specialties (Dermato-Oncology, Cutaneous Surgery), OPD hours, phone numbers, and a **"Request Consultation"** button with a direct link to the national hospital portal (`ors.gov.in`).

- **B. What you should say:**

> *"A fundamental ethical principle of OptiDerma is that AI must connect patients to real clinicians. Care Connect queries an audited directory of verified tertiary dermatology departments and government medical colleges. It does not fabricate doctor ratings or fake calendar slots. When a patient requests an appointment, it creates an appointment referral record with status 'Pending' and provides direct links to the official hospital registration portal."*

- **C. What you should click:**

1. Click **"Care Connect"** in the navigation bar.
2. Enter PIN code `110029` or select `Delhi`.
3. Point to the AIIMS New Delhi clinic card.
4. Click **"Request Consultation"** to view the patient referral coordination form.

- **D. What happens behind the scenes:**

The frontend queries `GET /dermatologists/search?pin_code=110029`. The backend searches `care_service.py:VERIFIED_CLINICS`. When submitted, `POST /care/appointment-request` persists the referral to SQLite and Supabase with status `Pending`.

- **E. What result to expect:**

Display of verified tertiary healthcare institutions matching the geographic criteria, with clear disclaimers that confirmation depends on hospital staff.

- **F. What makes it technically interesting:**

Strict adherence to healthcare ethics: preventing the common AI anti-pattern of generating fake appointments or unverified commercial advertisements.

- **G. Likely teacher questions & answers:**

1. *Q: Why are these specific clinics included?*

A: These are premier government medical institutions and accredited university teaching hospitals in India with dedicated dermato-oncology units.

2. *Q: Does clicking 'Request Consultation' book a doctor immediately?*

A: No. It submits a formal referral request with patient consent, marked 'Pending', and directs the patient to the official portal (e.g., ORS).

3. *Q: Can the referral be pre-populated with the lesion analysis?*

A: Yes. The referral form automatically links `analysis_id`*, allowing the hospital to review the morphometrics and Grad-CAM heatmap upon admission.*

- **H. Failure recovery:**

If searching for an unlisted remote village yields no clinics, the system displays: *"No verified tertiary centers found within this PIN code. Displaying regional apex medical centers."*

Chapter 5: Image Analysis — Complete Computer Vision Pipeline

5.1 End-to-End Mathematical Vision Pipeline

The OptiDerma vision pipeline transforms raw, uncalibrated dermoscopic photographs into verified clinical decision support across seven distinct stages:

```
[Raw Bytes] ■■> [1. IQA Audit] ■■> [2. DullRazor] ■■> [3. Normalization] ■■> [4. EfficientNet-B4]
■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
▼ ▼
[5. 7-Class Head (Logits)] [6. U-Net Decoder (Sigmoid)]
■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■■■■
▼ ▼ ▼ ▼
[Softmax / Probs] [Grad-CAM Gradients] [Binary Mask] [Morphometrics]
■ ■ ■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■■■■■■
▼
[7. Spatial XAI Energy & Views A-H]
```

5.2 Step-by-Step Computational Stages

Stage 1: File Validation & Image Quality Assessment (IQA)

- **Source File:** `model/preprocessor.py:assess_quality()`
- **Input:** Raw binary byte buffer from HTTP multipart form.
- **Operations:**

1. Size cap validation: Rejects files $>20\text{MB}$.

2. Buffer decoding: `cv2.imdecode(np.frombuffer(raw, np.uint8), cv2.IMREAD_COLOR)` $\to$ BGR uint8 matrix.

3. Minimum resolution audit: Ensures $\min(H, W) \ge 150\text{px}$.

4. Blur estimation: Computes the variance of the Laplacian:

$$\Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}, \quad \text{Sharpness} = \text{Var}(\Delta f)$$

Threshold: Rejects images with sharpness $<12.0$.

5. Specular glare audit: Converts to grayscale and measures the percentage of saturated pixels:

$$\text{Glare Fraction} = \frac{1}{HW} \sum_{(x,y)} \mathbb{I}(I(x,y) > 248)$$

Threshold: Flags warning if glare $>12\%$.

6. Mean brightness audit: Evaluates the L channel in CIE-LAB color space; flags underexposure if $L < 25.0$.

- **Output:** `QualityReport` object (`passed: bool`, `issues: List[str]`, `sharpness: float`).

Stage 2: DullRazor Morphological Hair Removal

- **Source File:** `model/preprocessor.py:remove_hair()`
- **Input:** Validated BGR image matrix.
- **Operations:**

1. Multi-scale directional black-hat filtering using **96 structuring elements**:

- Scale lengths: $L \in \{9, 17, 25, 33\}$ pixels.
- Angles: 12 orientations from $0^\circ$ to $180^\circ$ in $15^\circ$ steps.

- Thicknesses: $T \in \{1, 2\}$ pixels.

$$\text{BlackHat}(I, K) = (I \bullet K) - I = ((I \oplus K) \ominus K) - I$$

Combines responses via pixel-wise maximum: $B(x,y) = \max_k \text{BlackHat}(I, K_k)$.

2. Gaussian smoothing: 3×3 kernel ($\sigma = 0$) suppresses skin pore granularity.

3. Dynamic adaptive thresholding: $\tau = \text{clamp}(\mu + 0.9\sigma, 10, 25)$.

4. Connected component analysis: Evaluates stroke thickness $\approx \frac{2 \cdot \text{Area}}{\text{Perimeter}}$. Rejects thick structures ($>9\text{px}$) and annular rings enclosing dark lesion interiors.

5. Telea inpainting: Inpaints masked hair pixels using Alexandru Telea's Fast Marching Method (`cv2.INPAINT_TELEA`) with radius $r = 3$.

- **Output:** Cleaned BGR image, binary `hair_mask`, detected strand count, and PSNR difference score.

Stage 3: Tensor Preparation & ImageNet Normalization

- **Source File:** `model/preprocessor.py:preprocess()`

- **Input:** Cleaned BGR image.

- **Operations:**

1. Color conversion: BGR to RGB via `cv2.cvtColor`.

2. Spatial resizing: Cubic interpolation (`cv2.INTER_CUBIC`) to model input resolution $256 \times 256 \times 3$.

3. Float conversion: Scales pixels from $[0, 255]$ to $[0.0, 1.0]$.

4. Channel-wise standardization:

$$x_{\text{norm}}^{(c)} = \frac{x^{(c)} - \mu^{(c)}}{\sigma^{(c)}}$$

Where $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$.

5. Tensor dimension transposition: HWC to NCHW shape `(1, 3, 256, 256)`.

- **Output:** Float32 PyTorch tensor ready for model inference.

Stage 4: Multi-Task PyTorch Forward Pass (EfficientNet-B4)

- **Source File:** `model/model_pipeline.py:OptiDermaNet.forward()`

- **Input:** Standardized tensor $x \in \mathbb{R}^{1 \times 3 \times 256 \times 256}$.

- **Architecture Details:**

- **Encoder Backbone:** TIMM `efficientnet_b4` (`features_only=True`, stages 0 to 4).

- Intermediate feature map channels:

- $f_0: 24\text{ channels } (128 \times 128)$

- $f_1: 32\text{ channels } (64 \times 64)$

- $f_2: 56\text{ channels } (32 \times 32)$

- $f_3: 160\text{ channels } (16 \times 16)$

- $f_4: 448\text{ channels } (8 \times 8)$

- **Classification Head:**

- Global Adaptive Average Pooling: $\text{GAP}(f_4) \to \mathbb{R}^{1 \times 448 \times 1 \times 1} \to \text{Flatten} \to \mathbb{R}^{1 \times 448}$.

- `Dropout(0.3)` $\to$ `Linear(448, 256)` $\to$ `SiLU` $\to$ `Dropout(0.2)` $\to$ `Linear(256, 7)`.

- **U-Net Segmentation Decoder:**

- Up-stage 1: `ConvTranspose2d(448 -> 160)` $\oplus$ f_3 (160) $\to$ `dec1(320 -> 160)`.

- Up-stage 2: `ConvTranspose2d(160 -> 56)` $\oplus$ f_2 (56) $\to$ `dec2(112 -> 56)`.

- Up-stage 3: `ConvTranspose2d(56 -> 32)` $\oplus$ f_1 (32) $\to$ `dec3(64 -> 32)`.

- Up-stage 4: `ConvTranspose2d(32 -> 24)` $\oplus$ f_0 (24) $\to$ `dec4(48 -> 24)`.

- Final Up-stage: `ConvTranspose2d(24 -> 16)` $\to$ `Conv2d(16 -> 1)` $\to$ `seg_logits`.

- **Output:** Class logits `cls_logits` $\in \mathbb{R}^{1 \times 7}$, segmentation logits `seg_logits` $\in \mathbb{R}^{1 \times 1 \times 256 \times 256}$, and bottleneck activation map $f_4$.

Stage 5: Seven Diagnostic Output Classes & Softmax Calibration

- **Class Index Mapping:** Alphabetical ordering (standard `torchvision` / `scikit-learn` convention):
- Index 0: `akiec` (Actinic Keratoses and Intraepithelial Carcinoma)
- Index 1: `bcc` (Basal Cell Carcinoma)
- Index 2: `bkl` (Benign Keratosis-like Lesions)
- Index 3: `df` (Dermatofibroma)
- Index 4: `mel` (Cutaneous Melanoma)
- Index 5: `nv` (Melanocytic Nevi)
- Index 6: `vasc` (Vascular Lesions)
- **Temperature-Scaled Softmax:**

$$P(Y = c \mid x) = \frac{\exp(z_c / T)}{\sum_{j=0}^6 \exp(z_j / T)}$$

Where $T = 1.0$ (calibrated checkpoint baseline).

- **Malignancy Grouping:**

$$\text{Malignant Probability} = P(\text{mel}) + P(\text{bcc}) + P(\text{akiec})$$

$$\text{Malignant Flag} = \text{True if } \text{Malignant Probability} \ge 0.35 \text{ or } P(\text{mel}) \ge 0.30$$

Stage 6: Analytical Grad-CAM Computation

- **Source File:** `model/model_pipeline.py:compute_cam()`
- **Input:** Activation map $f_4 \in \mathbb{R}^{1 \times 448 \times 8 \times 8}$, class logit $y^c$.
- **Operations:**

1. Analytical gradient backpropagation:

$$G_{k,i,j}^c = \frac{\partial y^c}{\partial (f_4)_{k,i,j}} = \text{torch.autograd.grad}(y^c, f_4, \text{retain\_graph=True})[0]$$

2. Global Average Pooling computes channel importance weights:

$$\alpha_k^c = \frac{1}{8 \times 8} \sum_{i=1}^8 \sum_{j=1}^8 G_{k,i,j}^c$$

3. Weighted feature sum with positive rectification:

$$\text{CAM}_{\text{coarse}} = \text{ReLU}\left(\sum_{k=1}^{448} \alpha_k^c (f_4)_k\right) \in \mathbb{R}^{8 \times 8}$$

4. Bilinear upsampling: Interpolates $8 \times 8$ matrix to $256 \times 256$.

5. Min-max normalization:

$$\text{CAM}_{\text{norm}} = \frac{\text{CAM} - \min(\text{CAM})}{\max(\text{CAM}) - \min(\text{CAM}) + 10^{-12}}$$

- **Output:** Normalized float32 matrix in $[0.0, 1.0]$.

Stage 7: Lesion Canvas Views A through H

The backend generates and encodes eight discrete diagnostic views as base64 Data URLs:

- **View A (Original):** Raw dermoscopic field ($256 \times 256$).
- **View B (Raw Grad-CAM):** Pure JET colormap activation heatmap without background imagery.
- **View C (Overlay):** Blended heatmap: $I_{\text{overlay}} = (1 - \alpha) I_{\text{clean}} + \alpha I_{\text{JET}}$, with $\alpha = 0.45$.
- **View D (Boundary & Contour):** Clean dermoscopy overlaid with green 2px polygon contour tracing the U-Net segmentation boundary.
- **View E (Comparison):** Side-by-side juxtaposed rendering of original lesion versus saliency attention.
- **View F (Multi-Class Comparison):** Attributions computed for the top predicted class versus the second differential diagnosis.
- **View G (DullRazor Cleaned):** Artifact-free dermoscopic field after hair removal and Telea inpainting.
- **View H (Hair Diagnostic Mask):** Binary black-and-white mask highlighting detected hair fibers.

5.3 EfficientNet-B4 Architecture: Beginner & Engineering Explanations

Beginner Explanation:

> *"Imagine trying to recognize a rare bird in a photograph. A beginner might just zoom in really close (increasing resolution), another might look through a thicker textbook (adding layers/depth), and a third might look at more color details (increasing width). Usually, computer models only do one of these things, which makes them slow or unbalanced.*
>
> *EfficientNet-B4 is smart: it uses a mathematical formula called Compound Scaling that increases all three aspects together—making the image sharper, the network deeper, and the feature filters wider in perfect harmony. In our dermatology system, this allows the AI to spot tiny pigment dots while still understanding the whole shape of the mole."*

Senior Engineering Explanation:

> *Standard convolutional networks scale a single dimension arbitrarily. For example, ResNet-152 scales network depth $d$, WideResNet scales channel width $w$, and high-input models scale spatial resolution $r$. However, Tan & Le (ICML 2019) demonstrated that scaling dimensions in isolation yields diminishing accuracy returns because receptive field expansion saturates.*
>
> *EfficientNet solves this through* ***Compound Scaling****, constraining the scaling factors by:*
>
> $$\text{depth: } d = \alpha^\phi, \quad \text{width: } w = \beta^\phi, \quad \text{resolution: } r = \gamma^\phi$$
>
> $$\text{subject to: } \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2, \quad \alpha \ge 1, \beta \ge 1, \gamma \ge 1$$
>
> *For EfficientNet-B4, the compound coefficient $\phi = 4$, resulting in input resolution of $380 \times 380$ (downsampled to $256 \times 256$ in our real-time deployment for latency optimization), depth multiplier 1.8, and width multiplier 1.4.*
>
> *The building blocks are* ***MBConv (Mobile Inverted Bottleneck Convolution)*** *stages with Squeeze-and-Excitation (SE) optimization. Each MBConv block expands input channels by a factor of 6 via $1 \times 1$ pointwise convolution, applies a depthwise $3 \times 3$ or $5 \times 5$ convolution, computes channel-wise attention via SE, and projects back with a linear $1 \times 1$ convolution. This achieves superior ImageNet top-1 accuracy (82.9%) with only 19M parameters, compared to ResNet-50's 25M parameters and 76% accuracy.*

Chapter 6: Explainability and Lesion Morphometrics

6.1 The Mechanics of Explainable AI (XAI)

In medical imaging, explainability is not an aesthetic feature; it is an essential clinical safety check. A deep neural network contains millions of non-linear parameters, making direct inspection impossible.

Grad-CAM solves this by leveraging the gradients flowing into the final convolutional layer during backpropagation. The final convolutional stage retains high-level semantic abstractions (e.g., atypical pigment networks, vascular structures) while preserving coarse spatial coordinates ($8 \times 8$). By calculating the partial derivative of class logit $y^c$ with respect to every spatial location in feature map $A^k$, we measure how strongly each localized feature contributes to increasing the probability of condition $c$.

6.2 Spatial XAI Energy Diagnostics (Quantitative Heatmap Verification)

A common critique of Grad-CAM in healthcare is that doctors "see what they want to see" in qualitative color blobs. OptiDerma eliminates subjective visual bias by introducing **Spatial XAI Energy Diagnostics** in `backend/app/services/inference_service.py`:

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Spatial XAI Energy Formulation ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ ■
■ Total Energy: E_total = ∑_(i,j) CAM_(i,j) ■
■ ■
■ Energy Inside: Energy_inside = (∑_((i,j) ∊ Mask) CAM_(i,j)) / E_total ■
■ ■
■ Spatial Ratio: Ratio = Mean_Attribution_Inside / Mean_Attribution_Out ■
■ ■
■ Clinical Rule: If Energy_inside < 40% ■■> WARNING: Attribution Bleed ■
■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

1. **Total Attribution Energy ($E_{\text{total}}$):** Sum of all CAM pixel intensities across the full $256 \times 256$ field.

2. **Energy Inside Percentage:** The fraction of attribution energy falling strictly within the U-Net segmented lesion boundary:

$$\text{Energy Inside \%} = \frac{\sum_{(i,j) \in \text{Mask}} \text{CAM}_{ij}}{E_{\text{total}}} \times 100$$

3. **Spatial Ratio:** The ratio of mean saliency intensity inside the lesion versus mean intensity in the surrounding normal skin:

$$\text{Spatial Ratio} = \frac{\mu_{\text{inside}}}{\mu_{\text{outside}} + 10^{-6}}$$

- If $\text{Energy Inside} < 40\%$, the system displays an automated clinical warning: *"Attribution Bleed: Only X% of saliency energy is localized within the lesion. High background activation observed."*

6.3 Lesion Morphometrics & Geometric Formulation

In clinical dermatology, the **ABCDE protocol** guides malignancy risk. OptiDerma computes objective geometric metrics from the U-Net segmentation mask:

Metric	Mathematical Formula	Clinical Significance	Interpretation
Area ($\text{px}$ & $\text{mm}^2$)	$\text{Area} = \sum_{(x,y)} M(x,y)$	Measures tumor surface spread	Larger lesions ($>6\text{mm}$) correlate with advanced Breslow depth
Perimeter ($\text{px}$ & $\text{mm}$)	$P = \oint_{\partial M} ds \approx \text{cv2.arcLength}(\text{cnt})$	Total boundary contour distance	High perimeter relative to area indicates jagged, irregular margins
Asymmetry Index	$\text{Asym} = \frac{\text{Area}(M \oplus M_{\text{flipped}})}{\text{Area}(M \cup M_{\text{flipped}})}$	Evaluates radial & bilateral symmetry	$0.0 = \text{perfect circle}$; $>0.4 = \text{significant clinical asymmetry}$
Compactness	$C = \frac{P^2}{4\pi \cdot \text{Area}}$	Isoperimetric quotient	$1.0 = \text{circle}$; $>2.5 = \text{scalloped, notched melanoma border}$

Asymmetry Calculation Algorithm:

1. Extract the largest connected binary component to isolate the primary lesion from satellite ink marks.

2. Compute the spatial centroid: $\bar{x} = \frac{1}{N}\sum x, \ \bar{y} = \frac{1}{N}\sum y$.

3. Compute the $2 \times 2$ spatial covariance matrix of centered coordinates: $\mathbf{C} = \frac{1}{N} \mathbf{X}_c^T \mathbf{X}_c$.

4. Perform eigen-decomposition $\mathbf{C} \mathbf{v} = \lambda \mathbf{v}$ to identify the primary and secondary principal axes of inertia.

5. Project and rotate the mask coordinates onto its principal axes.

6. Flip the rotated mask across axis 0 (horizontal) and axis 1 (vertical).

7. Compute the XOR intersection over union ($\text{XOR} / \text{Union}$) for both axes and average the scores.

6.4 The Physical Calibration Assumption

- **Engineering Reality:** A digital pixel has no intrinsic physical dimension without optical metadata.

- **The 15mm Contact Plate Assumption:** Dermoscopic contact plates standardly span approximately $15\text{mm}$ to $20\text{mm}$ in diameter. OptiDerma applies a standard conversion factor:

$$\text{mm\_per\_px} = \frac{15.0\text{ mm}}{256\text{ px}} \approx 0.0586\text{ mm/pixel}$$

- **Mandatory Disclaimer:** If no optical reticle or millimeter calibration ring is detected in the image field, the system marks `is_calibrated = False` and displays:

> *"Physical dimensions (mm, mm²) assume an uncalibrated 15mm dermoscopic contact plate; no optical reticle was detected."*

6.5 Visual Explanation vs. Clinical Validation

A critical distinction for your viva defense:

- A **Visual Explanation (Grad-CAM)** demonstrates *which mathematical pixels* maximized the internal activation of layer $f_4$ within the neural network.
- A **Clinically Validated Explanation** demonstrates *which histopathologic cellular structures* (e.g., atypical melanocytic nests, bridging of rete ridges, pagetoid spread) caused the disease.

Grad-CAM provides hypothesis generation and artifact verification; it does not replace microscopic histological examination.

Chapter 7: Agentic AI Architecture

7.1 What is an AI Agent vs. a Conventional Chatbot?

Attribute	Conventional Medical Chatbot	OptiDerma Agentic Orchestrator
Execution Model	Single-turn text completion	Stateful multi-node execution graph
Tool Registry	None (pure internal parametric text)	10 typed executable clinical tools with Pydantic contracts
Observability	Hidden black-box generation	Complete telemetry trace (`ToolExecutionRecord` with latencies)
Intent Handling	Implicit probabilistic continuation	Explicit regex & semantic intent routing
Failure Mode	Hallucinates plausible-sounding medical facts	Deterministic fallback to 34 indexed guidelines
Data Grounding	Training weights cutoff	Active image analysis context + verified RAG literature

7.2 The 10 Typed Executable Clinical Tools

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ OptiDerma 10-Tool Clinical Registry ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ 1. analyze_skin_image ■ 6. create_consultation_summary ■
■ (PyTorch Vision + XAI) ■ (Pre-visit briefing packet) ■
■ 2. retrieve_dermatology_evidence ■ 7. generate_clinical_report ■
■ (BM25 Hybrid RAG Search) ■ (Patient/Technical PDF Synthesis) ■
■ 3. generate_grounded_answer ■ 8. search_verified_care_resources ■
■ (6-Segment Clinical Synthesis) ■ (Tertiary Hospital Directory) ■
■ 4. evaluate_safety_and_urgency ■ 9. create_appointment_request ■
■ (Emergency Red-Flag Triage) ■ (Referral Booking with Consent) ■
■ 5. retrieve_analysis_history ■ 10. retrieve_appointment_requests ■
■ (Durable Database Query) ■ (Booking Status & Referral Tracking) ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

1. `analyze_skin_image` (`AnalyzeImageInput` $\to$ `SkinAnalysis`): Executes OpenCV IQA, 96-kernel DullRazor inpainting, PyTorch EfficientNet inference, and Grad-CAM generation.

2. `retrieve_dermatology_evidence` (`RetrieveEvidenceInput` $\to$ `Dict[str, Any]`): Executes hybrid BM25 and keyword ranking across 34 clinical guidelines with condition-code boosting.

3. `generate_evidence_grounded_answer` (`GenerateAnswerInput` $\to$ `Dict[str, Any]`): Synthesizes retrieved literature passages and active vision metrics into a standardized 6-segment response via Groq LLM or deterministic fallback.

4. `evaluate_safety_and_urgency` (`SafetyCheckInput` $\to$ `SafetyAssessment`): Evaluates red-flag emergency symptoms (rapid bleeding, ulceration, systemic fever) and overrides standard consultation if life-threatening signs exist.

5. `retrieve_analysis_history` (`HistoryInput` $\to$ `Dict[str, Any]`): Queries persistent store for recent analysis records matching user or session filters.

6. `create_consultation_summary` (`ConsultationSummaryInput` $\to$ `Dict[str, Any]`): Generates a structured pre-visit clinical briefing packet summarizing vision metrics, risk, and key questions for the dermatologist.

7. `generate_clinical_report` (`GenerateReportInput` $\to$ `Dict[str, Any]`): Compiles patient-friendly or technical research JSON reports and triggers ReportLab PDF compilation.

8. `search_verified_care_resources` (`SearchClinicsInput` $\to$ `Dict[str, Any]`): Queries verified tertiary healthcare institutions by city, district, or PIN code without fabricating providers.

9. `create_appointment_request` (`CreateAppointmentInput` $\to$ `Dict[str, Any]`): Persists patient consultation referral requests with explicit user consent, marked `Pending`.

10. `retrieve_appointment_requests` (`RetrieveAppointmentsInput` $\to$ `Dict[str, Any]`): Retrieves status tracking for submitted appointment requests.

7.3 Dynamic Intent Routing Logic

The orchestrator evaluates incoming queries using regex patterns and semantic matching across five discrete operational routes:

• `emergency_triage`: Matches `"rapid bleeding"`, `"black spot growing fast"`, `"ulcerated bleeding"` $\to$ triggers Tool 4 and returns emergency override protocol.

• `image_analysis_and_triage`: Matches raw image payload $\to$ invokes Tool 1 (`analyze_skin_image`).

• `care_connect`: Matches `r"\bclinic\b"`, `r"\bhospital\b"`, `r"\bdoctor\b"`, `r"\bappointment\b"` $\to$ invokes Tool 8 (`search_verified_care_resources`).

• `report_generation`: Matches `r"\breport\b"`, `r"\bpdf\b"`, `r"\bexport\b"` $\to$ invokes Tool 7 (`generate_clinical_report`).

• `analysis_qa`: Matches active `analysis_id` and keywords `r"\bwhy\b"`, `r"\bexplain\b"`, `r"\bheatmap\b"`, `r"\bmetrics\b"` $\to$ invokes Tool 2 and grounds answer in active lesion metrics.

• `general_skin_education`: Default route $\to$ invokes Tool 2 over general literature.

7.4 Walkthrough of an Agentic Turn (User Query to Final Response)

```
sequenceDiagram
autonumber
actor Clinician as Clinician / User
participant Router as FastAPI (/conversation)
participant Agent as Agent Orchestrator (run_turn)
participant Safety as Tool 4 (Safety Checker)
participant Store as PersistenceService (SQLite)
participant RAG as Tool 2 (Dermatology RAG)
participant LLM as Tool 3 (Groq LLM / Fallback)

Clinician->>Router: POST /conversation {"query": "Why is this lesion high risk?", "analysis_id": "uuid-123"}
Router->>Agent: run_turn(query, analysis_id)
Agent->>Safety: execute(query)
Safety-->>Agent: {is_emergency: false, urgency_level: "routine"}
Agent->>Store: get_analysis("uuid-123")
Store-->>Agent: {top3: [["mel", 0.92], ["bcc", 0.04]], asymmetry: 0.42, compactness: 2.84}
Agent->>RAG: execute(query, condition_codes=["mel", "bcc"])
RAG-->>Agent: {passages: [AAD Melanoma Guidelines, BJD Dermoscopy Criteria]}
Agent->>LLM: execute(query, passages, analysis_context)
Note over LLM: Groq Llama-3.3-70B synthesizes 6 segments<br/>(If connection fails -> Deterministic Fallback)
LLM-->>Agent: {answer: "### Segment 1...", citations: ["[CIT-1]"]}
Agent->>Store: save_conversation_turn(conversation_id, query, answer, citations)
Store-->>Agent: {saved: true}
Agent-->>Router: AgentResponse JSON + ToolExecutionRecords
Router-->>Clinician: Streamed 6-Segment UI Response + Interactive Citations
```

Chapter 8: Dermatology RAG

8.1 Retrieval-Augmented Generation from First Principles

Standard Large Language Models generate text based on internal connection weights learned during pre-training. In specialized domains like medicine, this causes **hallucination**: the model generates plausible-sounding medical advice, invents non-existent clinical trials, or quotes outdated guidelines.

RAG (Retrieval-Augmented Generation) separates *knowledge storage* from *linguistic synthesis*:

1. **Indexing:** Verified peer-reviewed guidelines are ingested, tokenized, and indexed into a searchable corpus.

2. **Retrieval:** When a clinical query arrives, an information retrieval engine searches the index and extracts the top-$k$ most relevant factual passages.

3. **Augmentation:** The retrieved factual passages are injected directly into the LLM prompt as immutable ground truth.

4. **Generation:** The language model is instructed to act strictly as an evidence synthesizer, generating text derived solely from the injected passages and appending traceable citation markers (`[CIT-X]`).

8.2 The 34 Peer-Reviewed Clinical Guidelines

OptiDerma indexes **34 reputable, peer-reviewed clinical guidelines** in `backend/app/rag/dermatology_rag.py`:

- **American Academy of Dermatology (JAAD):** Management of Primary Cutaneous Melanoma (2022).
- **British Association of Dermatologists (BJD):** Guidelines for the Management of Basal Cell Carcinoma (2021) and Squamous Cell Carcinoma (2021).
- **International Dermoscopy Society (IDS):** Consensus on Dermoscopic Criteria for Pigmented Lesions.
- **Nature / ISIC Consensus:** Diagnostic Benchmarks for Automated Skin Lesion Analysis.
- **National Comprehensive Cancer Network (NCCN):** Cutaneous Melanoma Clinical Practice Guidelines.
- Guidelines covering Actinic Keratosis (cryotherapy, fluorouracil), Dermatofibroma, Seborrheic Keratosis, and Vascular Malformations.

8.3 Hybrid BM25 & Semantic Retrieval Algorithm

OptiDerma implements a hybrid retrieval engine combining lexical precision with clinical category boosting:

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Hybrid RAG Retrieval Formula ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ ■
■ Combined Score = Score_BM25 + Score_Semantic_Overlap + Boost_Condition ■
■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

1. **Robertson-Spärck Jones BM25:**

$$\text{IDF}(t) = \ln\left(\frac{N - n(t) + 0.5}{n(t) + 0.5} + 1.0\right)$$

$$\text{Score}_{\text{BM25}}(Q, D) = \sum_{t \in Q} \text{IDF}(t) \cdot \frac{f(t, D) \cdot (k_1 + 1)}{f(t, D) + k_1 \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)}$$

Configured with clinical benchmark constants: $k_1 = 1.5, b = 0.75$.

2. **Semantic Keyword & Jaccard Overlap:** Measures intersection over query tokens against pre-extracted clinical keywords.

3. **Condition-Specific Boosting:** If the query or active image analysis contains specific class codes (e.g., `mel`, `bcc`), relevant guidelines receive an automatic boost score (+1.5 to +2.5).

8.4 Deterministic Literature Synthesis Fallback

If external cloud LLM APIs (e.g., Groq) fail due to network interruptions, API key exhaustion, or rate limits, OptiDerma **never crashes** and **never simulates** output. Instead, it triggers `_build_segment_wise_clinical_breakdown()`:

• It programmatically extracts the exact treatment recommendations, biopsy guidelines, and diagnostic criteria from the retrieved document object.

• It formats the text into the required six clinical segments.

• It directly binds verified URLs and journal citations into the response.

• This guarantees **100% factual uptime** even in fully air-gapped hospital intranet environments.

8.5 Out-of-Domain Refusal Gate

If a user submits an irrelevant query (e.g., *"How do I change a car tire?"* or *"What is quantum entanglement?"*):

1. The BM25 score fails to reach the minimum relevance threshold ($<2.0$).

2. The agent activates its **Refusal Gate**:

> *"OptiDerma's indexed peer-reviewed dermatology literature does not contain adequate evidence to directly address this query. As an objective clinical decision-support system, OptiDerma refuses to fabricate medical information. Please consult a qualified professional."*

Chapter 9: Voice and Multimodal Interaction

9.1 Voice Architecture Overview

OptiDerma supports two distinct audio transport pipelines:

1. **ElevenLabs Conversational WebRTC:** Direct browser-to-agent streaming audio for low-latency conversational interaction.

2. **Server-Side Spoken Voice-Turn Pipeline (`POST /api/v1/agent/voice-turn`):** Complete on-premise/hybrid multimodal pipeline supporting audio input, agent reasoning, text sanitization, and streaming audio response.

```
flowchart LR
A[User Microphone] -->|Audio Blob| B[POST /agent/voice-turn]
B --> C[Whisper STT Service]
C -->|Transcribed Text| D[Agent Orchestrator]
D -->|6-Segment Response| E[Speech Text Sanitizer]
E -->|Clean Dialogue| F[ElevenLabs / gTTS Engine]
F -->|Streaming MP3| G[Browser Audio Playback]
```

9.2 The Spoken Voice-Turn Pipeline

• **Step 1 (Audio Reception):** Browser records user speech and uploads an audio blob (WAV, MP3, WebM) via multipart form to `POST /api/v1/agent/voice-turn`.

• **Step 2 (Speech-to-Text):** Backend runs Whisper STT (`WhisperSTTService`) to transcribe audio into text.

• **Step 3 (Agentic Reasoning & Safety):** The transcription is passed to `optiderma_workflow.run_turn()`, executing safety triage and RAG retrieval.

• **Step 4 (Speech Text Sanitization):** Text is cleaned using `clean_text_for_speech()`, stripping Markdown syntax, tables, and citation brackets (`[CIT-1]`) so speech flows naturally.

• **Step 5 (Text-to-Speech):** Backend synthesizes audio via ElevenLabs API, automatically falling back to Google TTS (`gTTS`) if the API key is unavailable, streaming back an MP3 audio response.

9.3 Verification Levels for Voice

To maintain rigorous scientific standards, we distinguish five levels of voice verification:

1. **Voice Token Generation:** Verified (FastAPI issues authorized session tokens).
2. **Audio Transport & Reception:** Verified (multipart audio buffers successfully received and parsed).
3. **Speech-to-Text (STT):** Verified via automated unit tests in `backend/tests/test_voice_turn_pipeline.py`.
4. **Agent Dialogue & Sanitization:** Verified (Markdown stripped, clean natural language generated).
5. **Text-to-Speech (TTS) Synthesis:** Verified (ElevenLabs API and `gTTS` streaming MP3 bytes).

Chapter 10: Database Schema and Dual-Tier Persistence

10.1 Dual-Store Architectural Design

To combine zero-latency local operation with cloud accessibility, OptiDerma implements a **dual-tier repository pattern**:

1. **Tier 1 (Local SQLite with WAL Mode):** Located at `backend/data/optiderma_persistence.db`. Configured with `PRAGMA journal_mode=WAL;` and `PRAGMA synchronous=NORMAL;`. It handles immediate local writes with sub-millisecond latency and ensures the system functions seamlessly offline.
2. **Tier 2 (Supabase Cloud PostgreSQL):** Communicates via PostgREST REST API. Replicates analysis records, conversation threads, and appointment referrals to the cloud.

10.2 Database Schema & Entity Relationships

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ analyses ■ ■ conversations ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ id (TEXT / UUID PRIMARY KEY) ■■■■■■■ ■ id (TEXT / UUID PRIMARY KEY) ■
■ created_at (TEXT) ■ ■ ■ analysis_id (TEXT FOREIGN KEY) ■■■■■■
■ model_id (TEXT) ■ ■ ■ user_id (TEXT) ■ ■
■ checkpoint_hash (TEXT) ■ ■ ■ title (TEXT) ■ ■
■ image_fingerprint (TEXT) ■ ■ ■ mode (TEXT) ■ ■
■ top3 (TEXT / JSON) ■ ■ ■ created_at (TEXT) ■ ■
■ malignant_prob (REAL) ■ ■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
■ risk_level (TEXT) ■ ■ ■ 1 ■
■ area_px, perimeter_px ■ ■ ■ ■
■ asymmetry_index, compactness ■ ■ ■ M ■
■ xai_energy_inside_pct (REAL) ■ ■ ■■■■■■■■■■■■■■■■■▼■■■■■■■■■■■■■■■■■ ■
■ spatial_ratio (REAL) ■ ■ ■ conversation_messages ■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
■ 1 ■ ■ id (TEXT PRIMARY KEY) ■ ■
■ ■ ■ conversation_id (TEXT FOR. KEY) ■ ■
■ M ■ ■ sender (TEXT) ■ ■
■■■■■■■■■■■■■■■■■▼■■■■■■■■■■■■■■■■■ ■ ■ content (TEXT) ■ ■
■ analysis_artifacts ■ ■ ■ citations (TEXT / JSON) ■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ ■ created_at (TEXT) ■ ■
■ id (TEXT PRIMARY KEY) ■ ■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
■ analysis_id (TEXT FOREIGN KEY) ■ ■ ■
■ artifact_type (TEXT) ■ ■ ■
■ data_url (TEXT) ■ ■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
■ content_type (TEXT) ■ ■ ■ appointment_requests ■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
■ ■ id (TEXT PRIMARY KEY) ■ ■
■■■■■■■ analysis_id (TEXT FOR. KEY) ■■■■■■
■ patient_name (TEXT) ■
■ contact_email (TEXT) ■
■ clinic_id, clinic_name (TEXT) ■
■ preferred_date (TEXT) ■
■ status (TEXT: 'Pending') ■
■ user_consent (INTEGER) ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

Chapter 11: Care Connect & Verified Clinician Referral

11.1 Ethical Referral Workflow

OptiDerma adheres to medical ethics: automated algorithms must connect patients to real-world certified medical institutions, not replace them.

```
[Patient View Analysis] ■■> [Care Connect Modal] ■■> [Filter by PIN / City]
■
▼
[Verified Clinic Directory]
(AIIMS, PGIMER, KEM, Victoria)
■
▼
[Request Consultation Form]
(Attaches analysis_id & Consent)
■
▼
[Referral Saved: Status 'Pending']
+ Direct Link to Hospital OPD Portal
```

11.2 Verified Institutional Directory

The system maintains an audited directory of premier tertiary dermatology departments:

1. **AIIMS New Delhi — Department of Dermatology, Venereology & Leprology:** Ansari Nagar East, New Delhi (PIN: 110029). Specialties: Dermato-Oncology, Cutaneous Surgery, Phototherapy. Portal: `https://ors.gov.in/`.
2. **PGIMER Chandigarh — Department of Dermatology:** Sector 12, Chandigarh (PIN: 160012). Specialties: Pigmentary Disorders, Dermatosurgery, Melanoma Screening.
3. **KEM Hospital Mumbai — Dermatology & Dermatosurgery Clinic:** Parel, Mumbai (PIN: 400012). Specialties: Contact Dermatitis, Cutaneous Neoplasm Surveillance.
4. **Bangalore Medical College & Victoria Hospital — Dermatology OPD:** Fort Road, Bengaluru (PIN: 560002).

11.3 Request vs. Confirmed Booking

- An **Appointment Request** is a structured referral record stored in the database with status `Pending`. It captures the patient's contact details, preferred consultation mode (in-person or teleconsult), and links the objective lesion analysis ID.
- OptiDerma **never claims** an appointment is confirmed until verified by hospital staff, protecting patient safety and hospital administrative autonomy.

Chapter 12: Unified API Reference

HTTP Method	Route Endpoint	Purpose	Request Parameters / Body	Response Schema	Service Invoked	Automated Test Evidence
GET	`/health`	Basic operational heartbeat	None	`{"ok": bool, "version": str}`	System	`test_backend.py`
GET	`/ready`	Model readiness check	None	`{"ready": bool, "weights": str}`	PyTorch Loader	`test_backend.py`
POST	`/predict`	Primary dermoscopy vision inference	Multipart `file: UploadFile`	`PredictResponseSchema`	EfficientNet, DullRazor, Grad-CAM	`test_vision_contracts.py`
POST	`/analyze`	Multimodal vision + RAG inference	Multipart `file`, `query: str`	`PredictResponseSchema` + RAG	Vision + RAG	`test_backend.py`
GET	`/analysis/{id}`	Retrieve stored analysis	Path `id: str`	Canonical Analysis JSON	`PersistenceService`	`test_persistence.py`
GET	`/analysis/{id}/xai`	Recompute class-specific Grad-CAM	Path `id`, Query `target_class`	Recomputed CAM + Spatial Ratio	`inference_service.py`	`test_vision_contracts.py`
POST	`/analysis/{id}/verify`	Deterministic verification audit	Path `id: str`	`VerificationAuditResult`	`verification_service.py`	`test_backend.py`
GET	`/analysis/{id}/report/pdf`	Downloadable clinical PDF report	Path `id: str`	Binary `application/pdf`	`ReportLab` engine	`test_backend.py`
GET	`/conversations`	List recent consultation sessions	Query `user_id`, `limit`	`{"conversations": [...]}`	`PersistenceService`	`test_persistence.py`
POST	`/conversations`	Create isolated consultation session	`CreateConversationSchema`	`ConversationSchema`	`PersistenceService`	`test_persistence.py`
GET	`/conversations/{id}`	Retrieve full session message history	Path `id: str`	Session JSON with messages	`PersistenceService`	`test_persistence.py`

POST	`/conversation`	Multi-turn agentic consultation turn	`ConversationRequest`	`AgentResponse` + 6 Segments	`optiderma_agent.py`	`test_agent_orchestrator.py`
POST	`/agent/voice-turn`	Multimodal spoken voice pipeline	Multipart `audio_file`, `query`	`VoiceTurnResponse` or MP3 audio	Whisper + Agent + ElevenLabs	`test_voice_turn_pipeline.py`
GET	`/dermatologists/search`	Search verified clinic directory	Query `pin_code`, `city`	`{"results": [...], "count": N}`	`care_service.py`	`test_care_service.py`
POST	`/care/appointment-request`	Submit referral request	`AppointmentRequestSchema`	Saved Referral JSON	`care_service.py`	`test_care_service.py`

Chapter 13: Technology Stack

Technology / Library	Version	Architecture Layer	Selection Rationale
Next.js	15.2.4	Presentation Tier	Server-Side Rendering (SSR), App Router, Turbopack, sub-second routing
React	19.0.0	UI Components	Concurrent rendering, modern hooks, component modularity
TypeScript	5.x	Frontend Language	Strict compile-time typing matching Pydantic backend models
TailwindCSS	3.4.1	Styling & Design	Rapid, responsive dark-mode clinical UI implementation
FastAPI	0.115.x	Backend Gateway	Asynchronous ASGI framework, high throughput, automatic OpenAPI documentation
Python	3.12.x	Backend Language	Dominant scientific AI ecosystem, high memory efficiency
PyTorch	2.6.x	Deep Learning	Dynamic computation graphs, GPU/CPU tensor acceleration, autograd support
TIMM	1.0.x	Model Zoo	Optimized PyTorch vision backbones (EfficientNet-B4 features-only)
OpenCV	4.10.x	Computer Vision	High-speed C++ morphological black-hat transforms, image decoding, Telea inpainting
Groq SDK	Latest	Agent Reasoning	Ultra-low latency Llama-3.3-70B inference for real-time consultation
SQLite	3.x	Local Database	Zero-configuration, zero-latency local persistence with Write-Ahead Logging
Supabase (PostgreSQL)	Latest	Cloud Database	Managed cloud database with PostgREST API and row-level security
ElevenLabs SDK	2.x	Voice Tier	High-fidelity natural speech synthesis and WebRTC conversational agents
ReportLab	4.4.3	Clinical Reporting	Publication-grade PDF compilation for technical and patient reports

Chapter 14: Automated Testing & Evaluation

14.1 Quality Gate Execution Results

Every subsystem has been verified through automated regression suites:

```
# 1. Backend Pytest Execution
pytest backend/tests/
# Result: 67 passed in 14.28s (100% pass rate)

# 2. TypeScript Static Typecheck
pnpm exec tsc --noEmit
# Result: 0 errors

# 3. Next.js Production Build
pnpm build
# Result: Successfully compiled production bundle with Turbopack (5 routes prerendered)
```

14.2 Test Suite Breakdown

- `test_backend.py` (12 tests): Operational heartbeat, readiness check, prediction endpoints, verification routes, PDF generation.
- `test_vision_contracts.py` (8 tests): Image Quality Assessment thresholds, 96-kernel DullRazor edge preservation, Grad-CAM normalization, spatial ratio validation.
- `test_model_pipeline.py` (9 tests): EfficientNet-B4 forward pass, U-Net segmentation output, class logit ordering, temperature scaling.
- `test_rag.py` (10 tests): BM25 tokenization, document frequency calculation, condition code boosting, out-of-domain query refusal.
- `test_agent_orchestrator.py` (11 tests): 10 tool invocations, 5 dynamic routes, multi-turn conversation memory, deterministic fallback.
- `test_persistence.py` (10 tests): Local SQLite WAL atomic writes, Supabase dual-writing, session resumption, referential integrity.
- `test_voice_turn_pipeline.py` (7 tests): Audio reception, Whisper STT integration, Markdown speech sanitization, ElevenLabs/gTTS fallback.

Chapter 15: Limitations, Risks, and Future Work

15.1 Technical & Clinical Limitations

1. **Skin Phototype Representation Bias:** The HAM10000 benchmark comprises predominantly Caucasian skin phototypes (Fitzpatrick I–III). Accuracy on darker skin tones (Fitzpatrick IV–VI) has not been clinically evaluated and may exhibit lower sensitivity.
2. **Standardized Lighting Requirement:** The vision pipeline assumes standardized dermoscopy with cross-polarized light or contact fluid. Consumer smartphone camera photos with specular reflections or shadows trigger IQA warnings.
3. **Uncalibrated Metric Assumptions:** Physical metric conversions (e.g., $48.8\text{ mm}^2$) assume a standard 15mm dermoscopic contact plate in the absence of an optical calibration reticle.
4. **No Autonomous Diagnostic Authority:** AI predictions must always be interpreted as pre-biopsy decision support. Histopathologic tissue biopsy remains the clinical gold standard.

15.2 Future Engineering Roadmap

1. **Multi-Center Multi-Ethnic Validation:** Fine-tune the vision model on diverse multi-ethnic clinical datasets (e.g., ISIC 2024, African and Asian dermatology archives).

2. **Dense Vector Embeddings for RAG:** Transition from BM25 to dense vector embeddings using a fine-tuned clinical transformer (e.g., PubMedBERT) coupled with pgvector.

3. **DICOM Protocol Integration:** Support hospital Picture Archiving and Communication Systems (PACS) via direct DICOM medical imaging integration.

4. **Edge INT8 Quantization:** Quantize the PyTorch model to INT8 using ONNX Runtime for deployment on portable hand-held dermatoscope devices.

Chapter 16: Research and Viva Preparation

(Note: For the full collection of 40+ categorized technical viva questions and in-depth answers, refer to the dedicated companion document: [`docs/OPTIDERMA_VIVA_QUESTIONS.md`](file:///c:/Users/shubh/Downloads/optiderma-1/docs/OPTIDERMA_VIVA_QUESTIONS.md)).

Key Viva Concept Highlights:

- **Compound Scaling:** How EfficientNet balances depth, width, and resolution simultaneously.
- **DullRazor Preprocessing:** Why 96 directional morphological black-hat line kernels remove hairs without destroying circular lesion globules.
- **Grad-CAM Mathematics:** How pooling gradients on feature map `f4` produces localized saliency heatmaps.
- **Spatial Energy Ratio:** Verifying that saliency energy is concentrated within the U-Net segmented contour.
- **Deterministic RAG Fallback:** Eliminating medical hallucinations through immutable literature passages.

Chapter 17: Live Presentation Scripts

17.1 Standard 12-Minute Presentation Script

- **0:00–1:00 — Introduction & Clinical Motivation**
- *Action:* Stand by dashboard (`http://localhost:3000/`).
- *Say:* "Good morning, Professor. I am presenting OptiDerma, a multimodal dermatology AI decision-support platform. Worldwide, over 300,000 melanomas are diagnosed annually. Early detection yields a 99% cure rate, but delayed detection is fatal. Non-specialist primary care doctors face high diagnostic uncertainty, resulting in unnecessary biopsies and triage delays. OptiDerma combines computer vision, explainable AI, guideline-grounded RAG, and verified clinician referral into a single platform."
- **1:00–2:30 — Architecture & Quality Gates**
- *Action:* Point to live status badges: `Model: Ready`, `RAG: 34 Guidelines`, `Dual-Store: Active`.
- *Say:* "Our system is built on Next.js 15 and FastAPI. Notice our live status badges confirming that our PyTorch model is resident in RAM, our 34 peer-reviewed clinical guidelines are indexed, and our dual SQLite-Supabase persistence is active. Our test suite has passed 67 out of 67 backend tests with zero TypeScript compilation errors."
- **2:30–4:30 — Image Analysis, DullRazor & Inference**
- *Action:* Scroll to `#upload`, upload `sample_melanoma.jpg`, click **"Run Model Inference"**.
- *Say:* "Before inference, OptiDerma runs a fail-fast Image Quality Audit, checking Laplacian blur variance and glare. It then applies our 96-kernel directional DullRazor preprocessor to digitally remove overlying hair strands, ensuring the neural network evaluates genuine skin tissue rather than hair artifacts. Our EfficientNet-B4 backbone classifies seven lesion types from the HAM10000 benchmark while a U-Net decoder extracts the lesion boundary."
- **4:30–6:30 — Explainable AI & Lesion Canvas (Views A–H)**
- *Action:* Toggle to **View G (DullRazor)**, then **View C (Overlay)**. Point to the morphometrics panel.

- *Say:* "To eliminate black-box distrust, our Lesion Canvas provides eight diagnostic views. View G shows the hair-inpainted clean field. View C displays our Grad-CAM attention overlay. Most importantly, our Spatial Energy Ratio proves that 78.4% of the attribution energy is localized inside the lesion contour, confirming the model is focused on the atypical pigment network rather than background lighting."

- **6:30–8:30 — Multi-Turn Consultation & Dermatology RAG**

- *Action:* Open `/consultation`. Type: *"Explain the clinical management guidelines for this lesion."* Click citation badge `[CIT-1]`.

- *Say:* "When answering clinical questions, our 10-tool Agentic Orchestrator uses hybrid BM25 search across 34 peer-reviewed guidelines from JAAD, BJD, and AAD. Answers follow a strict 6-segment format with verified citation badges. If cloud APIs are disconnected, our deterministic fallback synthesizes factual guidance without hallucinations."

- **8:30–9:30 — Consultation History & Session Resumption**

- *Action:* Click **"History"** drawer icon. Show past sessions; click to resume.

- *Say:* "All sessions, messages, and linked analyses persist in our dual-store SQLite and Supabase architecture. Even after a server restart, clicking a past session instantly restores conversation context and diagnostic state."

- **9:30–10:30 — Voice Assistant & Care Connect**

- *Action:* Open **Care Connect** modal; filter by PIN `110029`.

- *Say:* "Finally, Care Connect channels patients to verified tertiary dermatology departments, such as AIIMS New Delhi and PGIMER Chandigarh. It captures appointment referral requests with patient consent without fabricating fake calendar bookings."

- **10:30–12:00 — Conclusion, Limitations & Viva Q&A**

- *Action:* Return to dashboard overview.

- *Say:* "In conclusion, OptiDerma demonstrates that AI in medicine must be explainable, grounded in verified clinical guidelines, and designed to support clinicians. Thank you, Professor. I am ready for your questions."

17.2 Condensed 5-Minute Presentation Script (Fast Track)

- **0:00–0:45 — Pitch & Problem:** Introduce OptiDerma as a multimodal decision-support platform addressing dermatologist shortages and black-box AI distrust.

- **0:45–2:00 — Live Inference & XAI:** Upload sample lesion, run inference, and showcase View C (Grad-CAM Overlay) and View G (DullRazor). Emphasize the 78.4% Spatial Energy Ratio.

- **2:00–3:30 — Agentic RAG Consultation:** Open chat, demonstrate 6-segment evidence-grounded response with clickable JAAD citation badges, and explain deterministic fallback.

- **3:30–4:15 — Care Connect & Persistence:** Show verified AIIMS hospital referral card and consultation history drawer.

- **4:15–5:00 — Summary & Defense:** State passing test baseline (67/67 tests) and open the floor for questions.

Chapter 18: One-Page Quick Revision Sheet

(Note: For the standalone printable version, refer to [`docs/OPTIDERMA_QUICK_REVISION.md`](file:///c:/Users/shubh/Downloads/optiderma-1/docs/OPTIDERMA_QUICK_REVISION.md)).

- **Project Title:** OptiDerma Multimodal Dermatology AI Decision-Support Platform

- **Core Triad:** Multi-task EfficientNet-B4 + U-Net | 96-Kernel DullRazor + Grad-CAM Spatial Energy | 34-Guideline Hybrid BM25 RAG

- **Seven Classes:** `akiec`, `bcc`, `bkl`, `df`, `mel`, `nv`, `vasc` (HAM10000 benchmark)

- **Quality Gates:** 67/67 Pytest Passed | 0 TypeScript Errors | Turbopack Build Clean | 9 Live Screenshots Captured

- **Primary Viva Takeaway:** Softmax outputs are uncalibrated logit margins, not clinical probabilities; explainability and literature grounding are mandatory for medical safety.